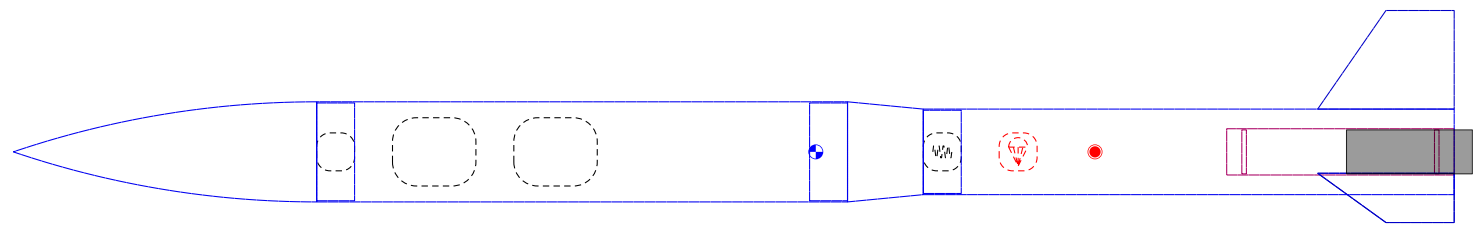


Rocket Design



800 Feet Rocket
Stages: 1
Mass (with motor): 438 g
Stability: 2.79 cal / 19.4 %
CG: 52.9 cm
CP: 71.3 cm

F23FJ-6

		Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
Altitude	244 m	F23FJ	18.5 N	2.2 s	27.5 N	40.7 Ns	4.30:1	30 g	29/83 mm
Flight Time	38.2 s								
Time to Apogee	7.51 s								
Optimum Delay	5.14 s								
Velocity off Pad	10.2 m/s								
Max Velocity	66.1 m/s								
Velocity at Deployment	7.95 m/s								
Landing Velocity	8.32 m/s								

Parts Detail

Stage: Sustainer

	Nose Cone	PLA - 100% infill (1.25 g/cm³)	Ogive	Len: 20 cm	Mass: 54.1 g
	Body Tube	Cardboard (0.68 g/cm³)	Dia _{in} 6.45 cm Dia _{out} 6.6 cm	Len: 35 cm	Mass: 36.6 g
	Egg 1		Dia _{out} 4.5 cm		Mass: 60 g
	Egg 2		Dia _{out} 4.5 cm		Mass: 60 g
	Ballast		Dia _{out} 2.5 cm		Mass: 18 g
	Transition	PLA - 100% infill (1.25 g/cm³)	Fore Dia: 6.6 cm Aft Dia: 5.64 cm	Len: 5 cm	Mass: 18.8 g
	Lower Body Tube	Cardboard (0.68 g/cm³)	Dia _{in} 5.49 cm Dia _{out} 5.64 cm	Len: 35 cm	Mass: 31.2 g
	Trapezoidal Fin Set (3)	PLA - 100% infill (1.25 g/cm³)	Thick: 0.16 cm		Mass: 26.3 g
	Inner Tube	Cardboard (0.68 g/cm³)	Dia _{in} 2.9 cm Dia _{out} 3.05 cm	Len: 15 cm	Mass: 7.15 g
	Centering Ring	Cardboard (0.68 g/cm³)	Dia _{in} 0 cm Dia _{out} 2.9 cm	Len: 0.3 cm	Mass: 1.35 g
	Centering Ring	Cardboard (0.68 g/cm³)	Dia _{in} 0 cm Dia _{out} 2.9 cm	Len: 0.3 cm	Mass: 1.35 g
	Parachute	Ripstop nylon (67 g/m²)	Dia _{out} 40 cm	Len: 2.5 cm	Mass: 13.8 g
	Shroud Lines	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)	Lines: 6	Len: 50 cm	
	Shock Cord	Tubular nylon (11 mm, 7/16 in) (13 g/m)		Len: 200 cm	Mass: 26 g

